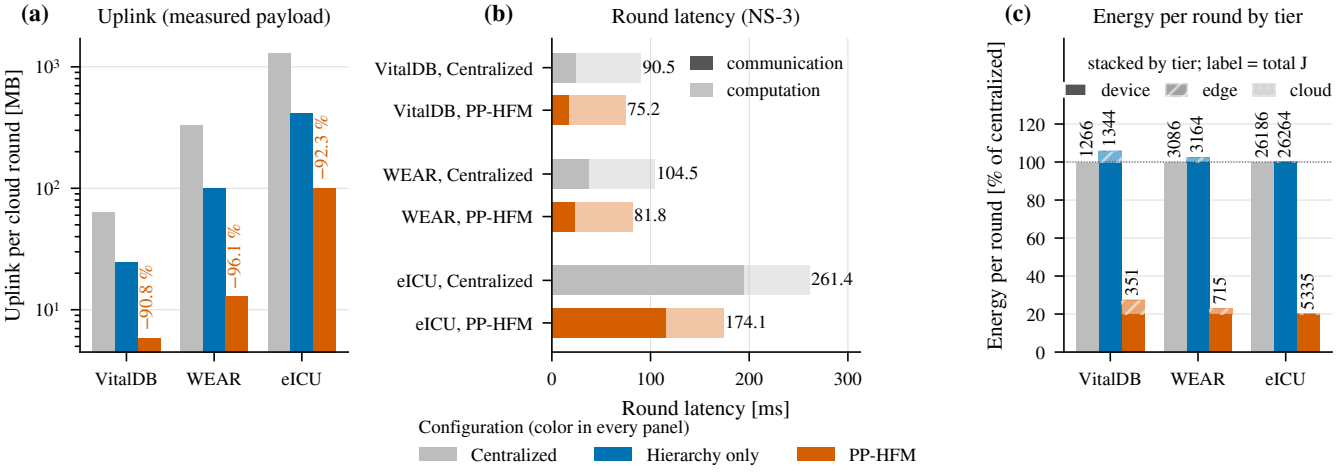

(a)
Uplink (measured payload)
Uplink per cloud round [MB]
10^3
10^2
10^1
−90.8 %
−96.1 %
−92.3 %
VitalDB
WEAR
eICU
(b)
Round latency (NS-3)
VitalDB, Centralized
VitalDB, PP-HFM
WEAR, Centralized
WEAR, PP-HFM
eICU, Centralized
eICU, PP-HFM
90.5
75.2
104.5
81.8
261.4
174.1
communication
computation
0
100
200
300
Round latency [ms]
(c)
Energy per round by tier
stacked by tier; label = total J
device
edge
cloud
Energy per round [% of centralized]
0
20
40
60
80
100
120
1266
1344
351
3086
3164
715
26186
26264
5335
VitalDB
WEAR
eICU
Configuration (color in every panel)
Centralized
Hierarchy only
PP-HFM